

Answer Script

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Answers:

Q1. $O(\log n)$

Q2. A list is mutable, meaning it can be changed after creation. A tuple is immutable and cannot be changed.

Q3.

```
def reverse_string(s):
```

```
    result = ""
```

```
    for char in s:
```

```
        result = char + result
```

```
    return result
```

Q4. Encapsulation is hiding data. Inheritance is subclassing. Polymorphism is method overriding.